

The effects of chronic exercise interventions on executive function in healthy older adults and optimal training characteristics: A systematic review based on randomized controlled trials

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ABSTRACT

In an aging society, mitigating cognitive decline is a critical challenge. However, reviews on exercise effects and optimal parameters for healthy older adults remain limited. This systematic review evaluated the effects of chronic exercise interventions on executive function in healthy older adults. Following PRISMA guidelines, we registered the review with PROSPERO (CRD42024618878) and searched Scopus, PubMed, EBSCOhost, and other sources for studies published before December 2024. We included randomized controlled trials involving adults aged ≥ 60 , with pre- and post-intervention executive function measures, comparing chronic exercise with non-exercise controls. Studies involving physical or neurological conditions or acute interventions were excluded. Risk of bias was assessed via Cochrane RoB 2 tool. Data synthesis followed the PICOS framework, with narrative and quantitative summaries. A total of 76 studies ($n = 7101$; 69 % female) met inclusion criteria. Of these, 71 % (54/76) showed significant improvements in executive function. Aerobic, resistance, and coordination exercises had similar effects (64 %–71 %). Optimal parameters included short session duration (≤ 45 min), high frequency (5–7/week), long-term duration (25–48 weeks), and vigorous intensity. High adherence appeared to enhance outcomes. These findings support chronic exercise as an effective strategy to improve executive function in healthy older adults and highlight the importance of exercise type, dosage, and adherence in promoting healthy aging.

1. Introduction

Population aging has become a pressing global issue, driven by increased life expectancy and the growing proportion of elderly individuals. According to the World Health Organization (WHO), the global population aged 60 and above exceeded one billion in 2020, accounting for approximately 13.5 % of the world's total population. This number is expected to rise significantly in the coming decades (World Health Organization, 2021). This demographic shift presents major challenges, particularly in terms of economic stability and healthcare system burdens (Grinin et al., 2023). The impact of aging on health is not only limited to physiological aspects but also profoundly affects cognitive function, especially the decline in executive functions, which directly influence quality of life and the ability to live

independently (Davis et al., 2010). Such cognitive decline increases risks, including accidents and exposure to dangerous situations, which can further jeopardize safety (Daigneault et al., 2002). Therefore, maintaining cognitive health has become a key strategy for promoting healthy aging, a priority for governments and public health organizations worldwide.

Executive function is a set of top-down high-level mental processes that enable goal-directed behavior, including three core components: working memory, inhibitory control, and cognitive flexibility (Diamond, 2020). Higher-level executive functions, such as reasoning, problem-solving, and planning, are built upon these core functions, and are crucial for managing daily tasks, making decisions, and navigating social situations (Diamond, 2020). Working memory involves maintaining and manipulating perceptual information, inhibitory control

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refers to regulating internal impulses and external distractions, and cognitive flexibility is the ability to adapt quickly in changing circumstances (Diamond, 2020). With aging, executive functions tend to decline, leading to difficulties in handling complex tasks, reducing independence, and increasing the risk of developing Alzheimer's disease, dementia, or other cognitive disorders (McGuinness et al., 2010). Deficits in executive function also contribute to poorer emotional regulation and social interactions, which can negatively impact overall well-being (Singh & Young, 2020). Therefore, understanding the factors that influence executive function in older adults is crucial for developing effective interventions to mitigate cognitive decline and promote healthy aging.

Chronic exercise, defined as structured and repeated physical activity undertaken over several weeks to induce long-term improvements in physical or cognitive functioning (Jasim et al., 2024), has been proposed as a promising intervention to enhance executive function in older adults by counteracting age-related cognitive decline through physical stimulation (Kraft, 2012). While exercise training has been shown to benefit executive functions, variations in its effects are thought to result from differences in the exercise prescriptions employed (e.g., type, duration, frequency, length, or intensity) (Chen et al., 2020). Therefore, further research is needed to identify the optimal exercise prescription to maximize cognitive benefits. Exercise type refers to the form of exercise, such as aerobic, resistance, coordination, or mixed exercise (which includes exergames and cognitive-motor training). Duration refers to the length of each exercise session, frequency is the number of sessions per week, length is the overall duration of the intervention (typically measured in weeks or months), and intensity refers to the effort level of the exercise, ranging from light to vigorous (American College of Sports Medicine, 2013). These parameters affect the outcomes of exercise interventions, emphasizing the need for tailoring exercise prescriptions based on individual needs and goals to achieve optimal physical and executive function health benefits.

While the importance of tailoring exercise prescriptions to maximize cognitive benefits is evident, current reviews on this topic remain insufficient. A previous review examined the effects of physical exercise on executive function in cognitively healthy older adults, demonstrating that regular exercise training can enhance executive function (Xiong et al., 2021). However, this review included only 25 studies, providing a limited evidence base and failing to investigate the specific influence of exercise parameters. Additionally, another review investigated the effects of exergames on cognitive function in healthy older adults (Torre & Temprado, 2022). The 23 included studies showed inconsistent results. The inconsistencies may stem from the inclusion of non-randomized controlled trials and the broad assessment of cognitive domains. Similarly, a recent review explored the effects of different types of physical exercise, including aerobic, resistance, coordination, and mixed exercise, on executive function in older adults, reporting positive effects in 70.4 %–83.3 % of studies (Cai et al., 2024). Despite these findings, this review included 22 studies involving individuals with mild cognitive impairment, limiting the generalizability of its conclusions to healthy older adults. Furthermore, although it evaluated the prevalence of various exercise parameters across studies, it did not determine which parameters are most effective for improving executive function. In addition, an overview of 18 review studies found that exergames and cognitive-motor training have positive effects on cognitive function. However, it also highlighted that adherence remains an underexplored factor (Gallou-Guyot et al., 2020). Adherence to exercise interventions is crucial for achieving cognitive benefits, as it directly affects intervention effectiveness (Mahmood et al., 2023). However, existing reviews rarely examine adherence or extract adherence rates to analyze their impact on executive function. Understanding this relationship is essential for optimizing intervention effectiveness. To address these gaps, comprehensive reviews are needed to clarify the effects of chronic exercise and identify the optimal parameters for enhancing executive function specifically in healthy older adults.

Under the framework of the United Nations Decade of Healthy Ageing (2021–2030), conducting an updated systematic review on how different exercise prescription parameters influence improvements in executive function among older adults is particularly timely. This effort is essential for incorporating recent research advancements, enhancing global data collection, and supporting national initiatives to promote healthy aging worldwide (Keating, 2022). Therefore, this systematic review aims to examine the effects of chronic exercise interventions on executive function in healthy older adults. To minimize the confounding effects of disease, this study focuses on healthy older individuals, avoiding the complications introduced by various health conditions that may complicate result interpretation (Chen et al., 2020). Additionally, this review emphasizes randomized controlled trials, as they are considered the gold standard for minimizing bias and establishing causal relationships between exercise interventions and cognitive outcomes (Bhide et al., 2018). By synthesizing the results of high-quality studies, this review aims to address three key questions: (1) What are the overall effects of chronic exercise interventions on executive function? (2) Which types of exercise are most effective in improving executive function? (3) How do specific exercise prescription parameters influence the effectiveness of chronic exercise interventions on executive function?

2. Methods

This systematic review was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Page et al., 2021) and registered with the International Prospective Register of Systematic Reviews (PROSPERO) under registration number: CRD42024618878. The full protocol is available at: <https://www.crd.york.ac.uk/PROSPERO/view/CRD42024618878>.

2.1. Search strategy

A comprehensive literature search was conducted across multiple electronic databases, including Scopus, PubMed, and EBSCOhost, using a combination of keywords and Boolean operators. The search terms included “physical activity” OR “exercise” OR “sport” OR “aerobic” OR “resistance” OR “coordination” OR “exergame” OR “training” AND “executive function” OR “executive control” OR “inhibitory control” OR “working memory” OR “updating” OR “cognitive flexibility” OR “switching” AND “elderly” OR “older adults” OR “old people” OR “old age” OR “aging” AND “randomized controlled trial” OR “randomized clinical trial” OR “controlled clinical trial.” To refine the results, the terms “acute” OR “bout” were excluded. Controlled vocabulary, such as Medical Subject Headings (MeSH) in PubMed, was used to improve search precision. The search was limited to studies published in English prior to December 2024. Additional relevant studies were identified by screening the reference lists of included articles and reviewing existing systematic reviews. The full search strategies for all databases, including search strings, filters, and limits, are provided in Appendix A.

2.2. Inclusion and exclusion criteria

Peer-reviewed studies were included if they met the following criteria: (1) involved at least one chronic exercise intervention of any type, along with a non-exercise control group or condition; (2) assessed executive function in a seated condition at both baseline and post-intervention using validated, performance-based cognitive tasks recognized as standardized neuropsychological assessments with established psychometric properties (e.g., N-back, Stroop, or Task Switching tasks); (3) included healthy older adults aged 60 or above, as defined by the WHO (World Health Organization, 2021); and (4) employed a randomized controlled trial design.

Studies were excluded if they (1) included participants with diagnosed physical and neurocognitive disorders; (2) focused on acute or

single-bout exercise interventions; (3) did not report executive function outcomes; (4) were non-peer-reviewed publications, such as conference abstracts, dissertations, or theses; or (5) were not written in English.

2.3. Study selection

Data extraction from the selected studies was conducted by two independent reviewers (CCL and TYC). In the first phase, studies retrieved from database searches were imported into EndNote v20, where duplicates were automatically removed using the built-in "Remove Duplicates" function. The remaining records were exported to Microsoft Excel for a secondary review to identify additional duplicates not detected by EndNote and to visually screen out records that did not meet the predefined inclusion criteria. In the second phase, titles and abstracts were screened to assess relevance and determine preliminary eligibility. In addition, supplementary searches were conducted using reference lists of previous systematic reviews and academic platforms such as Google Scholar to identify additional potentially relevant studies based on their titles and abstracts. In the third phase, full-text articles were retrieved and reviewed in detail to confirm inclusion. Studies that met the inclusion criteria at each stage progressed to the next. This stepwise screening process ensured that only eligible studies were included in the final analysis. Any disagreements during the selection process were resolved through discussion or consultation with a senior reviewer (TMH).

2.4. Risk of bias assessment

The quality and potential risk of bias of the included studies were thoroughly assessed using the revised Cochrane Risk of Bias Tool for Randomized Trials (RoB 2) (Sterne et al., 2019). The assessment followed established guidelines and evaluated five domains: (1) the randomization process, (2) deviations from intended interventions, (3) missing outcome data, (4) outcome measurement, and (5) selection of reported results. Each domain was rated as "low risk," "some concerns," or "high risk" based on specified criteria. Detailed guidance on the use of the tool is available at www.riskofbias.info. The assessment was conducted by two independent reviewers (CCL and TYC), with any discrepancies resolved through discussion or by consulting a senior reviewer (TMH).

2.5. Data extraction

Data extraction was conducted using a standardized form to ensure consistency across studies. The inclusion criteria were guided by the Population, Intervention, Comparison, Outcome, and Study Design (PICOS) framework (Amir-Behghadami & Janati, 2020) to maintain a rigorous and systematic selection process. Two independent reviewers (CCL and TYC) extracted the following information from the included articles: (1) study details, including first author, publication year, and country; (2) population characteristics, such as grouping, sample size, mean age, and sex ratio; (3) intervention details, including exercise type, duration, frequency, length, and intensity; (4) comparison measures for executive function, such as assessment tools for subcomponents like working memory, inhibitory control, and cognitive flexibility (excluding non-executive function outcomes); and (5) main outcomes included both within-subject findings (e.g., pre- and post-intervention changes within the experimental group) and between-subject findings (e.g., comparisons between the experimental and control groups at post-intervention), where reported. Adherence data were also extracted to provide a more comprehensive understanding of intervention implementation, with most studies defining adherence as attendance rates at scheduled sessions (Hawley-Hague et al., 2016). Adherence was categorized based on the median as follows: low adherence (<60 %), moderate adherence (60–80 %), or high adherence (>80 %) (Bailey et al., 2020). The study design was limited to randomized controlled

trials to ensure methodological rigor. Disagreements during the extraction process were addressed through discussion or by consulting with the senior reviewer (TMH).

Exercise intervention parameters were categorized following recent literature (Cai et al., 2024) and aligned with the FITT principle as described by the American College of Sports Medicine (Liguori, & American College of Sports Medicine, 2020). Exercise type was classified into four categories: aerobic, resistance, coordination, or mixed exercise (Ludyga et al., 2020). These classifications are based on their distinct effects on various physiological systems and are commonly used in exercise science to guide training programs. Duration was divided into three groups: short (≤ 45 min), medium (46–60 min), and long (≥ 61 min) (Northey et al., 2018). Frequency was grouped as low (1–2 sessions per week), moderate (3–4 sessions per week), and high (5–7 sessions per week) (Zhang et al., 2023). Intervention length was categorized into four terms: short-term (≤ 12 weeks), mid-term (13–24 weeks), long-term (25–48 weeks), and very long-term (≥ 49 weeks) (Zhang et al., 2023). Exercise intensity was categorized into five levels: very light, light, moderate, vigorous, and near-maximal to maximal based on %HRR or %VO₂R, %HRmax, %VO₂max, %1RM, or RPE metrics, in accordance with the American College of Sports Medicine (ACSM) Guidelines (Garber et al., 2011). If these indicators were not described, the reported exercise intensity in the study was used as the basis for classification.

To better characterize the existing literature, we categorized studies by exercise type and examined parameter interactions (e.g., duration, frequency, length, intensity) across different levels (e.g., low, moderate, high). This allows for an assessment of how exercise parameters interact in study distribution, highlighting research trends and potential gaps.

3. Results

3.1. Study selection

A total of 1279 records were identified through the initial database search. After removing duplicates and incomplete records, 624 studies remained for screening. Titles and abstracts were independently reviewed, resulting in the exclusion of 561 studies due to irrelevance, leaving 63 studies for further evaluation. In addition, supplementary searches using previous systematic reviews and Google Scholar identified 60 additional studies based on title and abstract screening. In total, 123 studies underwent full-text review and were assessed in detail against the predefined eligibility criteria. Ultimately, 76 studies met the inclusion and exclusion criteria and were included in the systematic review for subsequent analysis. The PRISMA flow diagram is shown in Fig. 1.

3.2. Results of the risk of bias assessment

Two independent reviewers (CCL and TYC) assessed the risk of bias in the included studies, achieving a high level of agreement (Cohen's Kappa = 0.865; McHugh, 2012).

As shown in Fig. 2, 35 studies had "low risk" and 41 had "some concerns" for bias arising from the randomization process. For bias due to deviations from intended interventions, 19 studies were rated as "low risk", 37 as "some concerns", and 20 as "high risk". Bias due to missing outcome data was identified as "low risk" in 44 studies, "some concerns" in 16, and "high risk" in 16. Regarding bias in outcome measurement, 19 studies were classified as "low risk" and 57 as "some concerns." Finally, 63 studies had "low risk", 12 had "some concerns," and one had "high risk" for bias in the selection of reported results.

3.3. Characteristics of studies and participants

Of the 76 studies included in this review, only one met the inclusion criteria prior to the year 2000 (Emery & Gatz, 1990). Between 2001 and 2010, nine studies were identified, followed by a substantial increase

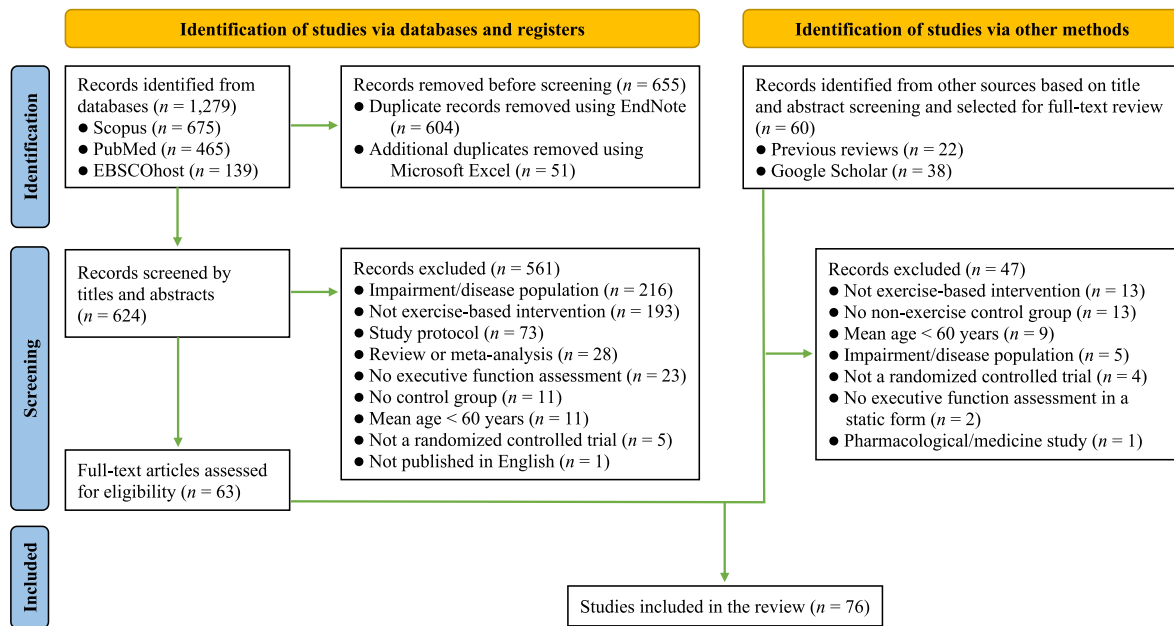


Fig. 1. PRISMA flow diagram of study selection.

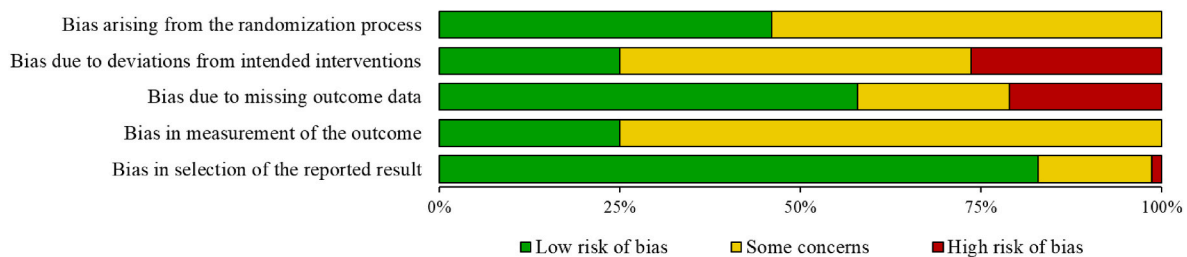


Fig. 2. Risk of bias summary (RoB 2).

Note. Risk levels: low risk (green), some concerns (yellow), and high risk (red).

between 2011 and 2020, with 48 studies meeting the criteria, accounting for 62.8 % of the total. After 2021, 18 studies were added.

Regarding geographic distribution, the United States contributed the highest number of studies ($n = 13$, 17.1 %). Brazil and Japan each contributed 9 studies, followed by Australia with 7. China and Germany each accounted for 6 studies, and Canada contributed 4. France, South Korea, and Taiwan each contributed 3 studies. Spain, Sweden, and Switzerland each provided 2 studies, while the Czech Republic, India, Iran, Italy, the Netherlands, Poland, and Portugal each contributed 1 study.

The studies collectively involved a total of 7101 participants. The average age of participants was 70.3 ± 4.4 years, with ages ranging from 64 to 85. In terms of sex, 2036 participants were male, representing 30.9 %, while 4548 were female, accounting for 69.1 %. Eight studies did not report information on sex. The summary of the characteristics of the studies and participants is provided in Table 1.

3.4. Intervention characteristics

Table 1 summarize the intervention characteristics across the included studies. Among the 76 studies, 15 included two different types of exercise intervention groups (e.g., aerobic and resistance exercise within the same study), and one study included three different types of exercise intervention groups (Iuliano et al., 2015), resulting in a total of 93 exercise groups. However, one group (Katayama et al., 2023) did not report the type of exercise, leaving 92 distinct exercise groups. Among these, aerobic exercise was the most frequently studied, appearing in 28

studies and accounting for 30.4 % of the 92 groups. This was followed by mixed exercises in 27 studies, and coordination exercises in 26 studies, while resistance exercises were the least frequently studied, appearing in only 11 studies.

Regarding session duration, three of the 76 studies did not report this parameter (Legault et al., 2011; Liu-Ambrose et al., 2012; Nouchi et al., 2014). However, two studies included two exercise groups with different session duration parameters (Oken et al., 2006; Taylor-Piliae et al., 2010). As a result, 75 studies were included in the subsequent analysis. Among these, most studies implemented a medium duration, reported in 41 of the 75 studies and accounting for 54.6 %. Shorter sessions were reported in 23 studies, while longer sessions were observed in 11 studies.

With respect to weekly session frequency, one study did not report this parameter (Legault et al., 2011), leaving 75 studies for further analysis. The most common weekly session frequency was moderate, reported in 44 of the 75 studies, representing 58.6 %. Low frequency, was reported in 28 studies, while high frequency was investigated in 3 studies (Cho & Roh, 2019; Ji et al., 2017; Seol et al., 2023).

The majority of intervention lengths were short-term, reported in 37 of the 76 studies, representing 48.6 %. This was followed by 30 studies with mid-term interventions and 8 studies with long-term interventions. Only one study implemented a very long-term intervention (Sink et al., 2015).

Exercise intensity was reported in 36 studies, including two that mentioned exercise intensity without detailing specific indicators (Sink et al., 2015; Williamson et al., 2009). Additionally, three studies

Table 1

Summary characteristics of the included studies.

Study (author, year, country)	Population (group, n, age, sex)	Intervention (type, duration, frequency, length, intensity); Adherence	Comparison (executive function tasks)	Outcome ($p < .05$)
Emery and Gatz (1990) United States	- Physical activity ($n = 14$, age = 72, NR) - Control ($n = 24$, age = 72, NR)	- Mixed: Stretching and aerobic (60 min, 3/week, 12 weeks, 70 % HRmax) - Adherence: 77.5 %	- Digit Symbol Test - DGS	No effect.
Oken et al. (2006) United States	- Yoga ($n = 44$, age = 71, M/F = 13/31) - Running ($n = 47$, age = 73, M/F = 10/37) - Control ($n = 44$, age = 71, M/F = 11/33)	- Coordination (90 min, 1/week, 6 months, NR) - Aerobic (60 min, 1/week, 6 months, 70 % HRR) - Adherence: 69 %	SCWT	No effect in both groups.
Cassilhas et al. (2007) Brazil	- Moderate-intensity ($n = 19$, age = 69, NR) - High-intensity ($n = 20$, age = 68, NR) - Control ($n = 23$, age = 67, NR)	- Resistance (60 min, 3/week, 24 weeks, Moderate: 50 % 1RM, High: 80 % 1RM) - Adherence: 75 %	CBTT	Both resistance groups had higher scores than the control group.
Brown et al. (2009) Australia	- Resistance and balance exercise ($n = 82$, age = 79) - Flexibility exercise ($n = 34$, age = 81) - Control ($n = 38$, age = 78) M/F = 19/135	- Mixed: Resistance and coordination, Coordination (60 min, 2/week, 6 months, NR) - Adherence: 63 %	- TMT (Part B) - SCWT - COWAT	No effect in both groups.
Marmeleira et al. (2009) Portugal	- Physical-cognition ($n = 16$, age = 68, M/F = 12/4) - Control ($n = 16$, age = 68, M/F = 13/3)	- Mixed: Dual-task activities, peripheral vision, planning and decision-making, speed processing, and response inhibition tasks (60 min, 3/week, 12 weeks, NR) - Adherence: NR	- TMT (Part B) - SCWT	No effect.
Williamson et al. (2009) United States	- Physical activity ($n = 50$, age = 76, M/F = 14/36) - Control ($n = 52$, age = 78, M/F = 16/36)	- Mixed: Aerobic, strength, balance, and flexibility (40–60 min, 3/week, 12 months, Moderate intensity) - Adherence: NR	- DSST - SCWT	No effect.
Kimura et al. (2010) Japan	- Resistance ($n = 65$, age = 73, M/F = 25/40) - Control ($n = 54$, age = 75, M/F = 24/30)	- Resistance (90 min, 2/week, 12 weeks, 60 % of 1RM) - Adherence: NR	Task Switching	No effect.
Klusmann et al. (2010) Germany	- Exercise ($n = 91$, age = 73) - Control ($n = 76$, age = 73) M/F = 0/167	- Mixed: Aerobic, strength, and flexibility (90 min, 3/week, 6 months, NR) - Adherence: NR	- TMT (Part A & B) - SCWT	No effect.
Liu-Ambrose et al. (2010) Canada	- Resistance 1 ($n = 54$, age = 69) - Resistance 2 ($n = 52$, age = 69) - Control ($n = 49$, age = 70) M/F = 0/155	- Resistance 1 (60 min, 1/week, 12 months, 6–8 × 2 reps, 7-RM) - Resistance 2 (60 min, 2/week, 12 months, 6–8 × 2 reps, 7-RM) - Adherence: 70.65 %	- SCWT - TMT (Part A & B) - DGS	Both resistance groups outperformed the control group on the SCWT (trial completion).
Taylor-Piliae et al. (2010) United States	- Tai Chi ($n = 37$, age = 70, M/F = 13/24) - Western exercise ($n = 39$, age = 68, M/F = 11/28) - Control ($n = 56$, age = 68, M/F = 15/41)	- Coordination (45 min, 4/week, 6 months, NR) - Mixed: Combines endurance, resistance/strength, and flexibility exercises (45–55 min, 4/week, 6 months, NR) - Adherence: 72.5 %	- Animal SFT - DGS	Tai Chi group showed better performance in the DGS compared to the control group.
Erickson et al. (2011) United States	- Walking ($n = 60$, age = 67, M/F = 16/44) - Control ($n = 60$, age = 65, M/F = 24/36)	- Aerobic (15–40 min, 3/week, 6 months, Target HRmax zone: 50–60 % for weeks 1–7, 60–75 % for remaining weeks) - Adherence: 79.5 %	SWM	The walking group showed improvement in SWM after the intervention.
Legault et al. (2011) United States	- Physical activity ($n = 18$, age = 77, M/F = 8/10) - Physical-cognition ($n = 19$, age = 76, M/F = 7/12) - Control ($n = 18$, age = 75, M/F = 11/7)	- Mixed: Aerobic and flexibility, Mixed: Physical-cognition (60 min/week, 4 months, NR) - Adherence: 83 %	- N-back - Flanker - Task Switching - TMT (Part A & B)	No effect in both groups.
Voelcker-Rehage et al. (2011) Germany	- Cardiovascular ($n = 17$, age = 68, M/F = 5/12) - Coordination ($n = 16$, age = 71, M/F = 6/10) - Control ($n = 11$, age = 69, M/F = 5/6)	- Aerobic, Coordination (60 min, 3/week, 12 months, - Cardiovascular: 35 min/session, increased to 45–50 min/session after ~3 months) - Adherence: 75 %	Flanker	Both exercise groups improved accuracy from pre- to post-test.
Evers et al. (2011) Germany	- Mixed-exercise ($n = 80$, age = 73) - Control ($n = 69$, age = 73)	Mixed: 30 min of cardiorespiratory endurance training followed by 60 min of strength,	- FCSRT - TMT (Part A & B)	The mixed-exercise group showed greater improvement in the composite

(continued on next page)

Table 1 (continued)

Study (author, year, country)	Population (group, n, age, sex)	Intervention (type, duration, frequency, length, intensity); Adherence	Comparison (executive function tasks)	Outcome ($p < .05$)
	M/F = 0/149	flexibility, balance, and coordination exercises (90 min, 3/week, 6 months, NR)		score of executive function from pre- to post-test compared to the control group.
Liu-Ambrose et al. (2012) Canada	● Resistance 1 ($n = 20$, age = 69) ● Resistance 2 ($n = 15$, age = 68) ● Control ($n = 17$, age = 69) M/F = 0/52	● Adherence: 69.1 % ● Resistance 1 (NR, 1/week, 12 months, 6–8 × 2 reps, 7-RM) ● Resistance 2 (NR, 2/week, 12 months, 6–8 × 2 reps, 7-RM) ● Adherence: 77.15 %	● Flanker	● The resistance 2 group outperformed the control group (reduced interference).
Maillot et al. (2012) France	● Exergame ($n = 16$, age = 73) ● Control ($n = 16$, age = 73) M/F = 5/27	● Mixed (60 min, 2/week, 12 weeks, NR) ● Included Wii games like Tennis, Boxing, Bowling, Soccer Headers, Ski Jump, Slalom, Marbles, Hula Hoop, and Trampoline. ● Adherence: 97.5 %	● TMT (Part A & B) ● SCWT ● LST ● WAIS DSST	● The exergame group outperformed the control group in the TMT, SCWT, and LST.
Maki et al. (2012) Japan	● Walking ($n = 75$, age = 71, M/F = 23/52) ● Control ($n = 75$, age = 72, M/F = 21/54)	● Aerobic (90 min, 1/week, 12 weeks, NR) ● Included 30 min of exercise and 60 min of group work. ● Adherence: 87.5 %	● Dual-Task ● TMT (Part A & B)	● No effect.
Mortimer et al. (2012) United States	● Tai chi ($n = 30$, age = 67, M/F = 11/19) ● Walking ($n = 30$, age = 67, M/F = 11/19) ● Control ($n = 30$, age = 68, M/F = 9/21)	● Coordination, Aerobic (50 min, 3/week, 40 weeks, NR) ● Adherence: NR	● WAIS DGS ● ROCF ● SCWT ● AVL T ● Category VFT ● TMT (Part A & B)	● Tai Chi improved in ROCF (delayed recall), AVL T, and TMT. ● The walking group showed improvement in the copying condition of the ROCF.
Nguyen and Kruse (2012) Germany	● Tai chi ($n = 48$, age = 69, M/F = 24/24) ● Control ($n = 48$, age = 68, M/F = 24/24)	● Coordination (60 min, 2/week, 6 months, NR) ● Adherence: 81.25 %	● TMT (Part A & B)	● The Tai Chi group showed greater improvement in TMT (Part A & B) compared to the control group at post-test.
Chapman et al. (2013) United States	● Aerobic ($n = 18$, age = 64, M/F = 5/13) ● Control ($n = 19$, age = 64, M/F = 5/14)	● Aerobic (60 min, 3/week, 12 weeks, 50–75 % VO_2max) ● Includes bike and treadmill. ● Adherence: 90 %	● TMT (Part A & B) ● SCWT ● DGS	● No effect.
Hariprasad et al. (2013) India	● Yoga ($n = 62$, age = 75, M/F = 26/36) ● Control ($n = 58$, age = 74, M/F = 22/36)	● Coordination (60 min, 3–4/week, 6 months, NR) ● Included loosening exercises, breathing techniques, physical postures, and meditation. ● Adherence: NR	● RAVLT ● RCF ● WAIS DGS ● COWAT ● SCWT ● TMT (Part A & B)	● The Yoga group outperformed the control group in RAVLT, RCF, WAIS DGS, COWAT, SCWT, and TMT.
Schoene et al. (2013) Australia	● Videogame ($n = 15$, age = 77, NR) ● Control ($n = 17$, age = 78, NR)	● Mixed (15–20 min, 2–3/week, 8 weeks, NR) ● Adherence: 91.6 %	● TMT (Part A & B)	● No effect.
Shatil (2013) United States	● Aerobic ($n = 31$, age = 81, M/F = 7/22) ● Physical-cognition ($n = 29$, age = 79, M/F = 9/20) ● Control ($n = 29$, age = 79, M/F = 12/19)	● Aerobic, Mixed (45 min, 3/week, 16 weeks, NR) ● Adherence: NR	● WCST ● SCWT	● The physical-cognition group outperformed the aerobic and control groups in both executive function tasks.
Kamegaya et al. (2014) Japan	● Physical activity ($n = 26$, age = 73, M/F = 2/24) ● Control ($n = 26$, age = 76, M/F = 3/23)	● Mixed: Stretching, strength training, and aerobic exercise (120 min, 1/week, 12 weeks, NR) ● Adherence: 77.6 %	● WAIS DSST ● YKSST	● The physical activity group outperformed the control group in the WAIS DSST.
Nouchi et al. (2014) Japan	● Combination exercise ($n = 30$, age = 66, NR) ● Control ($n = 31$, age = 67, NR)	● Mixed: Aerobic, strength, and stretching (NR, 3/week, 4 weeks, Resistance exercise: 60–80 % HRmax) ● Adherence: NR	● Stroop ● NIH VFT ● DGS	● The combination exercise group improved more in the Stroop and VFT than the control group in the post-test.
Vaughan et al. (2014) Australia	● Physical activity ($n = 25$, age = 69) ● Control ($n = 24$, age = 68) M/F = 0/49	● Mixed: Cardiovascular, strength, and motor fitness training (60 min, 2/week, 16 weeks, NR) ● Adherence: 85 %	● COAST ● COWAT ● LNS ● TMT (Part A & B)	● The physical activity group outperformed the control group in three tests (excluding LNS).
Ansai and Rebelatto (2015) Brazil	● Multicomponent ($n = 23$, age = 81, M/F = 6/17) ● Resistance ($n = 23$, age = 82, M/F = 8/15) ● Control ($n = 23$, age = 82, M/F = 8/15)	● Mixed: Aerobic, strength, and coordination exercises, Resistance (60 min, 3/week, 16 weeks, Aerobic: 60–85 % HRR) ● Resistance ● Adherence: 45.6 %	● VFT ● Dual-Task	● No effect in both groups.
Hanna Karen M Antunes et al. (2015) Brazil	● Cycle ergometer ($n = 22$, age = 68, NR) ● Control ($n = 23$, age = 66, NR)	● Aerobic (60 min, 3/week, 6 months, NR) ● Adherence: NR	● DGS Numbers ● ROCF ● WCST	● The cycle ergometer group improved in the ROCF (immediate recall), DGS forward, and WCST.

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Table 1 (continued)

Study (author, year, country)	Population (group, n, age, sex)	Intervention (type, duration, frequency, length, intensity); Adherence	Comparison (executive function tasks)	Outcome ($p < .05$)
Hanna Karen Moreira Antunes et al. (2015) Brazil	● Training ($n = 23$, age = 64) ● Control ($n = 17$, age = 64) M/F = 0/40	● Aerobic (60 min, 3/week, 6 months, NR) ● Adherence: NR	● DGS ● WCST ● ROCF	● The training group showed greater improvement than the control group in DGS forward, ROCF (immediate and delayed recall), and WCST.
H. Antunes et al. (2015) Brazil	● Cycle ergometer ($n = 23$, age = 68) ● Control ($n = 23$, age = 65) M/F = 46/0	● Aerobic (60 min, 3/week, 6 months, NR) ● Adherence: NR	● CBTT	● The cycle ergometer group improved scores from pre- to post-test. ● The cycle ergometer outperformed the control group in the post-test.
Ferreira et al. (2015) Brazil	● Walking ($n = 22$, age = 66, M/F = 7/15) ● Control ($n = 22$, age = 69, M/F = 3/19)	● Aerobic (40–50 min, 3/week, 6 months, 60–80 % HRR) ● Adherence: NR	● CBTT ● WCST	● The walking group outperformed the control group in categories completed and fewer perseverative errors on the WCST.
Nishiguchi et al. (2015) Japan	● Physical-cognition ($n = 24$, age = 73, M/F = 13/11) ● Control ($n = 24$, age = 73, M/F = 13/11)	● Mixed (90 min, 1/week, 12 weeks, NR) ● VFT during short, fast-step exercises. ● Adherence: 91.7 %	● TMT (Part A & B)	● The physical-cognition group had a shorter completion time than the control group.
Nocera et al. (2015) United States	● Spin exercise ($n = 10$, age = 72, M/F = 4/6) ● Control ($n = 8$, age = 71, M/F = 4/4)	● Aerobic: Stationary cycling exercises (45 min, 3/week, 12 weeks, 75 % HRR) ● Adherence: 83 %	● D-KEFS VFT	● Spin exercise outperformed the control group in the category condition.
Schoene et al. (2015) Australia	● Exergame ($n = 47$, age = 82, M/F = 16/31) ● Control ($n = 43$, age = 81, M/F = 14/29)	● Mixed (20 min, 3/week, 16 weeks, NR) ● Adherence: NR	● Stroop ● Stepping Test ● VST ● TMT (Part A & B) ● DSC ● ROCF ● TMT (Part A & B) ● Animal VFT ● Oddball Task	● No effect.
Sink et al. (2015) United States	● Physical activity ($n = 735$, age = 78) ● Control ($n = 741$, age = 78) M/F = 477/999	● Mixed: Aerobic, strength, flexibility, coordination (50 min, 3–4/week, 24 months, Walking at a moderate intensity) ● Adherence: 71 %	● DSC ● ROCF ● TMT (Part A & B) ● Animal VFT ● Oddball Task	● No effect.
Tsai et al. (2015) Taiwan	● Resistance ($n = 24$, age = 70) ● Control ($n = 24$, age = 72) M/F = 48/0	● Resistance (60 min, 3/week, 12 months, 75–80 % of 1RM) ● Adherence: 90 %	● SCWT ● TMT (Part A & B)	● The resistance group showed higher accuracy in the post-test compared to the control group. ● The resistance group reduced reaction time from pre- to post-test. ● All three exercise groups showed no effect.
Iuliano et al. (2015) Italy	● Resistance ($n = 20$, age = 65, M/F = 9/11) ● Cardiovascular ($n = 20$, age = 68, M/F = 8/12) ● Balance ($n = 20$, age = 66, M/F = 7/13) ● Control ($n = 20$, age = 66, M/F = 8/12)	● Resistance (30 min, 3/week, 12 weeks, 60–85 % of 1RM) ● Cardiovascular (30 min, 3/week, 12 weeks, 50–80 % HRR) ● Balance (30 min, 3/week, 12 weeks) ● Adherence: NR	● SCWT ● TMT (Part A & B)	● The swimming group outperformed the control group in SCWT. interference scores. ● The swimming group had more correct responses on the Running Span task at post-test.
Albinet et al. (2016) France	● Swimming ($n = 19$, age = 67, M/F = 6/13) ● Control ($n = 17$, age = 66, M/F = 4/13)	● Aerobic (60 min, 2/week, 21 weeks, 40–65 % HRR) ● Adherence: 80.3 %	● SCWT ● RNG ● Hayling Task ● Running Span Task ● N-back ● DST ● Plus-Minus Task ● Digit-Letter Task ● TMT (Part A & B)	● The swimming group outperformed the control group in SCWT. interference scores. ● The swimming group had more correct responses on the Running Span task at post-test.
Merom et al. (2016) Australia	● Dancing ($n = 279$, NR, M/F = 48/231) ● Control ($n = 251$, NR, M/F = 34/217)	● Coordination (60 min, 2/week, 12 months, NR) ● Folk dancing or ballroom dancing ● Adherence: 56 %	● TMT (Part A & B)	● No effect.
Ji et al. (2017) China	● Physical exercise ($n = 12$, age = 67, M/F = 7/5) ● Control ($n = 12$, age = 73, M/F = 5/7)	● Mixed: Wii program (30 min, 7/week, 6 weeks, NR) ● Included aerobic, balance, weightlifting, and yoga exercises with Nintendo ● Adherence: NR	● WAIS DSSM ● WAIS DGS ● TMT (Part A & B) ● SCWT ● Flanker ● OSPAN ● DGS ● Letter Memory Task ● N-back ● Keep Track Task ● TMT (Part A & B)	● The physical exercise group improved in the WAIS DSSM from pre- to post-test.
Jonasson et al. (2017) Sweden	● Aerobic ($n = 29$, age = 68, M/F = 14/15) ● Control ($n = 29$, age = 68, M/F = 12/17)	● Aerobic (30–60 min, 3/week, 6 months, 40–80 % HRmax) ● Walking, jogging, cycling, or using cross-trainers. ● Adherence: NR	● SCWT ● Flanker ● OSPAN ● DGS ● Letter Memory Task ● N-back ● Keep Track Task ● TMT (Part A & B)	● Aerobic outperformed the control group in the broad cognitive score.

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Table 1 (continued)

Study (author, year, country)	Population (group, n, age, sex)	Intervention (type, duration, frequency, length, intensity); Adherence	Comparison (executive function tasks)	Outcome ($p < .05$)
			● Odd-Even Task ● Local-Global Task	
Tsai et al. (2017) Taiwan	● Open-skill ($n = 22$, age = 66) ● Closed-skill ($n = 21$, age = 66) ● Control ($n = 21$, age = 65) M/F = 64/0	● Coordination (table tennis) ● Aerobic (bicycle ergometer: 70–75 % HRR) (40 min, 3/week, 24 weeks) ● Adherence: 90 %	● Task Switching ● N-back	● The open-skill group reduced reaction time on the Task Switching. ● Both groups improved accuracy on the N-back task: the open-skill group in 1-back and the closed-skill group in 1-back and 2-back.
Cherup et al. (2018) United States	● Circuit-resistance ($n = 9$, age = 72, NR) ● Treadmill ($n = 8$, age = 68, NR) ● Control ($n = 7$, age = 70, NR)	● Resistance, Aerobic (35 min, 3/week, 12 weeks) ● Resistance: 40–70 % of 1RM. ● Treadmill: 55 % HRR) ● Adherence: 83 %	● Flanker ● NIH DCCS ● NIH PSMT ● NIH LSWM ● NIH PCPST	● Resistance training improved PCPST performance from pre- to post-test. ● Both exercise groups improved Flanker performance from pre- to post-test.
Gajewski and Falkenstein (2018) Germany	● physical activity ($n = \text{NR}$, age = 70, NR) ● Control ($n = \text{NR}$, age = 70, NR)	● Mixed: Aerobic and strength exercises ● Adherence: NR	● ROCF ● DGS ● N-Back	● The physical activity group reduced reaction time in the 2-back task from pre- to post-test
Wu et al. (2018) Taiwan	Total sample ($n = 141$) ● Tai Chi Chuan ($n = 16$, age = 64, M/F = 3/13) ● Control ($n = 15$, age = 64, M/F = 0/15)	● Coordination (60 min, 3/week, 12 weeks, 63–66 % HRmax) ● Adherence: NR	● Number Stroop Task	● The Tai Chi Chuan group reduced errors in task-switching performance from pre- to post-test.
Bouaziz et al. (2019) France	● Interval aerobic ($n = 30$, age = 72, M/F = 9/21) ● Control ($n = 30$, age = 74, M/F = 7/23)	● Aerobic (30 min, 2/week, 9.5 weeks, 6 sets of 4 min at VT_1, followed by 1 min at 40 % of VT_1) ● Adherence: NR	● NIH VFT ● TMT (Part A & B)	● Interval aerobic better than the control group in reducing time on the TMT and VFT.
Cho and Roh (2019) Korea	● Taekwondo ($n = 19$, age = 68) ● Control ($n = 18$, age = 69) M/F = 0/37	● Coordination (60 min, 5/week, 16 weeks, 50–80 % HRmax) ● Adherence: NR	● SCWT	● The Taekwondo group had higher scores than the control group in the color-word condition.
Kleinloog et al. (2019) Netherlands	● Aerobic ($n = 8$, age = 67) ● Control ($n = 9$, age = 67) M/F = 17/0	● Aerobic (50 min, 3/week, 8 weeks, cycling ergometer: 70 % $VO_{2\text{max}}$) ● Adherence: 100 %	● MTT ● SSP	● The aerobic group reduced reaction time in the MTT.
Lopes Filho et al. (2019) Brazil	● Karate ($n = 16$, age = 69, M/F = 1/15) ● Control ($n = 17$, age = 68, M/F = 1/16)	● Coordination (60 min, 2/week, 12 weeks, NR) ● Adherence: NR	● TMT (Part A & B) ● DGS ● ROCF ● RAVLT ● WCST ● VFT ● N-back	● The Karate group outperformed the control group in RAVLT, ROCF recall, WCST, and TMT.
Norouzi et al. (2019) Iran	● Resistance-motor dual-task ($n = 20$, age = 68, M/F = 20/0) ● Resistance-cognition dual-task ($n = 20$, age = 68, M/F = 20/0) ● Control ($n = 20$, age = 68, M/F = 20/0)	● Mixed: Resistance training with simultaneous motor skill training. ● Resistance: resistance training with 12 simultaneous cognitive tasks. (60–80 min, 3/week, 12 weeks, NR) ● Adherence: NR		● Resistance-cognition group outperformed the other groups in scores of N-back from pre- to post-test.
Prehn et al. (2019) Germany	● Moderate aerobic ($n = 11$, age = 69, M/F = 7/4) ● Control ($n = 18$, age = 65, M/F = 8/10)	● Aerobic (45 min, 2/week, 6 months, cycling ergometer: 80 % $VO_{2\text{max}}$) ● Adherence: NR	● TMT (Part A & B) ● DGS ● SCWT ● VFT	● Moderate aerobic exercise outperformed the control group on the TMT (Part B) from pre- to post-test.
Fang et al. (2020) China	● High-speed circuit resistance ($n = 12$, age = 70, M/F = 5/7) ● Control ($n = 8$, age = 71, M/F = 4/4)	● Resistance (40–45 min, 3/week, 8 weeks, 40–65 % of 1RM) ● Adherence: NR	● NIH Flanker Task ● NIH DCCS ● NIH LSWM	● The high-speed circuit resistance group showed improvement in Flanker task scores from pre- to post-test.
Santos et al. (2020) Brazil	● Resistance ($n = 24$, age = 66, M/F = 8/16) ● Control ($n = 25$, age = 68, M/F = 10/15)	● Resistance (50–60 min, 3/week, 12 weeks; three sets of 10–15 RM per exercise) ● Adherence: NR	● SCWT ● DGS ● VFT	● Resistance training was shown to improve performance on the SCWT, DGS, and VFT.
Adcock et al. (2020) Switzerland	● Training ($n = 15$, age = 77, M/F = 5/10) ● Control ($n = 16$, age = 70, M/F = 10/6)	● Mixed: Tai Chi-inspired exercises, dancing, and step-based cognitive games (30–40 min, 3/week, 4 months, NR) ● Adherence: 70 %	● VST ● TMT (Part A & B) ● WMS	● Intervention group improved in the VST and WMS from pre- to post-test.
Arrieta et al. (2020) Spain	● Intervention ($n = 57$, age = 85, M/F = 15/42) ● Control ($n = 55$, age = 84, M/F = 18/37)	● Mixed: Individualized, progressive multicomponent exercise focusing on strength, balance, and walking (60 min, 2/week, 6 months, Resistance: 40–70 % of 1RM). ● Adherence: 84.9 %	● NIH VFT ● Animal SFT	● No effect.

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Table 1 (continued)

Study (author, year, country)	Population (group, n, age, sex)	Intervention (type, duration, frequency, length, intensity); Adherence	Comparison (executive function tasks)	Outcome ($p < .05$)
Esmail et al. (2020) Canada	- Dancing ($n = 12$, age = 68, M/F = 4/8) - Aerobic ($n = 15$, age = 67, M/F = 4/11) - Control ($n = 14$, age = 67, M/F = 2/12)	- Coordination, Aerobic (60 min, 3/week, 12 weeks, 70 % VO_2max) - Adherence: 91.02 %	- Dual-Task - N-back - Digit Stroop Task	Both training groups improved in executive and non-executive composite scores post-intervention.
García-Garro et al. (2020) Spain	- Pilates ($n = 55$, age = 69) - Control ($n = 52$, age = 66) M/F = 0/107	- Coordination (60 min, 2/week, 12 weeks, NR) - Adherence: 91.6 %	- NIH VFT - TMT (Part A & B)	- The Pilates group improved in VFT and TMT scores from pre- to post-test. - The Pilates group showed better executive function gains than the control group.
Yang et al. (2020) China	- Tai Chi Chua ($n = 13$, age = 66) - Control ($n = 13$, age = 65) M/F = 0/26	- Coordination (45 min, 3/week, 8 weeks, NR) - Adherence: NR	Flanker	The Tai Chi Chuan group showed reduced reaction time in the incongruent condition from pre- to post-test.
Brown et al. (2021) Australia	- High-intensity ($n = 33$, age = 70, M/F = 16/17) - Moderate-intensity ($n = 34$, age = 68, M/F = 16/18) - Control ($n = 32$, age = 68, M/F = 13/19)	- Aerobic (50 min, 2/week, 6 months, High-intensity: 11 cycling intervals, 1 min >80 % VO_2max, 2 min recovery at 30–40 % VO_2max, Moderate-intensity: 50–60 % VO_2max) - Adherence: 85.9 %	- TMT (Part B) - NIH VFT - NIH Flanker Task - NIH Set-shifting Task	No effect in both groups.
Frost et al. (2021) Australia	- High-intensity ($n = 33$, age = 70, M/F = 16/17) - Moderate-intensity ($n = 34$, age = 68, M/F = 16/18) - Control ($n = 32$, age = 68, M/F = 13/19)	- Aerobic (50 min, 2/week, 6 months, High-intensity: 11 cycling intervals, 1 min >80 % VO_2max, 2 min recovery at 30–40 % VO_2max, Moderate-intensity: 50–60 % VO_2max) - Adherence: 85.9 %	- TMT (Part A & B) - CogState N-back - NIH FLNK	No effect in both groups.
Hou and Li (2022) China	- Exergame ($n = 23$, age = 67, M/F = 5/18) - Control ($n = 21$, age = 67, M/F = 6/15)	- Coordination: Dance-based exergaming (30 min, 3/week, 12 weeks, NR) - Adherence: NR	- Digit Symbol Test - TMT (Part B) - SCWT - N-back - WAIS DGS	No effect.
Miyazaki et al. (2022) Japan	- Nordic walking ($n = 29$, age = 67, M/F = 22/7) - Dancing ($n = 30$, age = 67, M/F = 17/13) - Control ($n = 29$, age = 68, M/F = 23/6)	- Aerobic, Coordination (45 min, 3/week, 4 weeks, NR) - Adherence: 91.6 %	- FAB VFT - FAB CIT - Go/No-Go Test	Both exercise groups had greater gains in inhibitory control (Go/No-Go Test) than the control group.
Park (2022) Korea	- Moderate-intensity ($n = 28$, age = 67, M/F = 14/14) - Control ($n = 28$, age = 67, M/F = 14/14)	- Aerobic (40 min, 3/week, 5 weeks, Maintain a Borg rating of 12–14 (moderate intensity)) - Adherence: NR	- SCWT - TMT (Part B)	The moderate-intensity aerobic group outperformed the control group in SCWT and TMT performance post-intervention.
Zhao et al. (2022) China	- Exergame ($n = 21$, age = 65, M/F = 7/14) - Dancing ($n = 18$, age = 65, M/F = 2/16) - Control ($n = 16$, age = 65, M/F = 6/10)	- Mixed (75 min, 3/week, 12 weeks, 65–75 % HRmax) - Coordination - Adherence: NR	- N-back - SCWT	The exergame group outperformed both the dancing and control groups in response accuracy during the 1-back condition of the N-back task.
Hirase et al. (2023) Japan	- Intervention ($n = 40$, age = 72, M/F = 13/27) - Control ($n = 39$, age = 72, M/F = 12/27)	- Mixed: Strength and balance (60 min, 1/week, 24 weeks, NR) - Adherence: 65.2 %	DSST	No effect.
Katayama et al. (2023) Japan	- Exercise ($n = 36$, age = 74, M/F = 29/7) - Control ($n = 32$, age = 75, M/F = 23/9)	- NR (90 min, 1/session, 12 weeks, NR) - Adherence: NR	TMT (Part A & B)	No effect.
Lichtenstein et al. (2023) Switzerland	- Agility ($n = 25$, age = 70, M/F = 10/15) - Control ($n = 29$, age = 69, M/F = 11/18)	- Coordination (60 min, 2/week, 12 months, Average 71 % HRmax) - Adherence: 75 %	- Flanker - N-back	The agility group showed improved reaction time in the N-back task.
Phillips et al. (2023) United States	- Exergame ($n = 14$, age = 73, M/F = 8/6) - Exercise ($n = 13$, age = 73, M/F = 7/6) - Control ($n = 13$, age = 70, M/F = 7/6)	- Mixed: Featured games that combined physical and cognitive activities. - Mixed: Included low-impact cardio, strength training, balance exercises, and stretching. (60 min, 2/week, 10 weeks, NR) - Adherence: 95.25 %	- Mazes Test - TMT (Part B) - Tower of London	The exergame group showed improved overall executive function from pre- to post-test.
Seol et al. (2023) Japan	- Manual dexterity ($n = 28$, age = 72, M/F = 9/19) - Control ($n = 29$, age = 74, M/F = 9/20)	- Coordination (20 min, 7/week, 12 weeks, NR) - Adherence: 88.7 %	SCWT	The manual dexterity group showed improvement in the interference condition from pre- to post-test.

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Table 1 (continued)

Study (author, year, country)	Population (group, n, age, sex)	Intervention (type, duration, frequency, length, intensity); Adherence	Comparison (executive function tasks)	Outcome ($p < .05$)
Trombini-Souza et al. (2023) Brazil	- Balance ($n = 30$, age = 67, M/F = 4/26) - Control ($n = 30$, age = 66, M/F = 4/26)	- Coordination (60 min, 2/week, 24 weeks, NR) - Adherence: 75 %	- TMT (Part A & B) - SCWT	The balance group improved execution time in the TMT and SCWT from pre- to post-test.
Wang et al. (2023) China	- Tai Chi ($n = 40$, age = 60–69, NR) - Control ($n = 40$, age = 60–69, NR)	- Coordination (60 min, 4/week, 12 weeks, NR) - Adherence: NR	- SCWT - N-back - MOST	The Tai Chi group had lower SCWT and N-back scores than the control group after the intervention.
Welford et al. (2023) Sweden	- Yoga ($n = 27$, age = 72, M/F = 7/20) - Aerobic ($n = 29$, age = 73, M/F = 7/22) - Control ($n = 26$, age = 72, M/F = 5/21)	- Coordination, Aerobic (60 min, 3/week, 12 weeks, Light-to-moderate intensity on Borg RPE scale) - Aerobic includes cycling/spinning and dance-based exercises. - Adherence: 61.05 %	VFT	No effect in both groups.
Byun et al. (2024) Korea	- Cycling ($n = 40$, age = 68, M/F = 10/30) - Control ($n = 41$, age = 67, M/F = 10/31)	- Aerobic (30–50 min, 3/week, 12 weeks, 35 % VO_2max) - Adherence: 97.6 %	SCWT	- The cycling group decreased reaction time from pre- to post-test. - The cycling group had shorter reaction time than the control group in the post-test.
Silva et al. (2024) Canada	- Physical-cognition ($n = 41$, age = 72, M/F = 19/22) - Control ($n = 42$, age = 71, M/F = 19/23)	- Mixed (60 min, 3/week, 8 weeks, 15 min brisk walk at 13–14 Borg RPE scale) - Adherence: 90 %	- NIH Flanker Task - NIH DCCS	Physical-cognition improved Flanker congruent performance from pre- to post-test.
Wiśniowska et al. (2024) Poland	- Cognitive-motor training ($n = 21$, age = 71, M/F = 7/14) - Control ($n = 20$, age = 70, M/F = 4/16)	- Coordination (30 min, 3/week, 4 weeks, NR) - Adherence: NR	- WCST - DGS - VFT - SCWT	The cognitive-motor training group showed improved cognitive flexibility, inhibition, and post-test working memory compared to the other group.
Hola et al. (2024) Czech Republic	- Dance ($n = 25$, age = 69, M/F = 5/20) - Martial Arts ($n = 25$, age = 71, M/F = 3/22) - Control ($n = 27$, age = 70, M/F = 5/22)	- Aerobic, Coordination (45 min, 2/week, 12 weeks, NR) - Adherence: 76.6 %	- RAVLT - Digit Symbol Test - TMT (Part A & B)	The dance group showed greater improvement in the Trail Making Test compared to the control group.

Note. Working Memory: Auditory Verbal Learning Test (AVLT), Automated Operation Span (OSPAN), CogState N-back, Corsi Block-Tapping Test (CBTT), Digit Span (DGS), Keep Track Task, Letter Memory Task, Letter-Number Sequencing (LNS), N-back, NIH List Sorting Working Memory Test (LSWM), NIH Picture Sequence Memory Test (PSMT), Rey Auditory Verbal Learning Test (RAVLT), Rey Complex Figure Test (RCF), Rey Osterrieth Complex Figure Test (ROCF), Running Span Task, Spatial Span Task (SSP), Spatial Working Memory Task (SWM), WAIS Digit Span (DGS), WAIS Digit Symbol Coding Task (DSC), Wechsler Memory Scale (WMS) Digit and Spatial Span, Free and Cued Selective Reminding Test (FCSRT).

Inhibitory Control: California Older Adult Stroop Test (COAST), Digit Stroop Task, Eriksen Flanker Task (Flanker), FAB Conflicting Instructions Task (CIT), Go/No-Go Task, Hayling Sentence Completion Test (Hayling Test), NIH Flanker Inhibitory Control and Attention Test (FLNK), NIH Pattern Comparison Processing Speed Test (PCPST), Number Stroop Task, Oddball Task, Random Number Generation (RNG) Task, Stroop Color-Word Test (SCWT), Stroop Stepping Test, and The Stroop Color-Word Test-Victoria version (VST).

Cognitive Flexibility: Controlled Oral Word Association Test (COWAT), Digit Symbol Substitution Test (DSST), Digit Symbol Test, Digit-Letter Task, Dimension-Switching Task (DST), D-KEFS Verbal Fluency Test (VFT), Dual-Task, FAB Verbal Fluency Task (VFT), Letter Sets Test (LST), Local-Global Task, Mazes Test, More-Odd Shifting Task (MOST), Multitasking Test (MTT), NIH Dimensional Change Card Sort Test (DCCS), Odd-Even Task, Plus-Minus Task, Semantic Fluency Test (SFT), Set-shifting Task, Task Switching, Tower of London, Trail Making Test (TMT) (Part A & B), Verbal Fluency Task (VFT), WAIS Digit Symbol Substitution Modality Test (DSSM), WAIS Digit Symbol Substitution Test (DSST), Wisconsin Card Sorting Test (WCST), and Yamaguchi Kanji-Symbol Substitution Test (YKSST).

reporting different intensity levels for their second exercise group (Brown et al., 2021; Cassilhas et al., 2007; Frost et al., 2021), resulting in a total of 39 exercise intensity parameters. Moderate intensity was the most commonly used, reported in 21 of the 39 studies (6 with resistance exercises), accounting for 53.8 %. Vigorous intensity was used in 15 studies (5 with resistance exercises). Light intensity was reported in two studies (Jonasson et al., 2017; Welford et al., 2023), and very light intensity was reported in one study (Byun et al., 2024). No studies reported near-maximal to maximal intensity levels.

Of the 76 studies, 32 did not report adherence data. Among the 44 studies that provided adherence information, high adherence was the most common, reported in 24 of the 44 studies, accounting for 54.5 %, followed by moderate adherence in 18 studies and low adherence in only two studies (Ansai & Rebelatto, 2015; Merom et al., 2016).

3.5. Executive function assessments

Among the included studies, 46 assessed multiple cognitive tasks,

while 30 focused on a single task. Cognitive flexibility was the most frequently assessed domain, with 60 studies, representing 78.9 %, employing 26 different tasks. The Trail Making Test (TMT) was the most common, used in 35 studies, followed by the Verbal Fluency Task (VFT), which appeared in 13 studies. Working memory was evaluated in 38 studies, using 21 different tasks. The Digit Span (DGS) was the most frequently used, appearing in 15 studies, followed by the N-back task, used in 11 studies. Inhibitory control was assessed in 44 studies with 14 different tasks. The Stroop Task was the most commonly used, appearing in 31 studies, followed by the Flanker Task, used in 11 studies. Table 1 summarizes the tasks used for each study.

3.6. Effect of intervention on executive function

The findings revealed that chronic exercise interventions significantly improved executive function in healthy older adults in 54 of 76 studies, accounting for 71 %.

Regarding exercise type, aerobic, resistance, and coordination

exercises demonstrated similar benefits. Aerobic exercise was effective in 20 of 28 studies, accounting for 71.4 %. Resistance exercise showed significant effects in 7 of 11 studies, equivalent to 63.6 %, while coordination exercise demonstrated efficacy in 18 of 26 studies, representing 69.2 %. Mixed exercise showed significant effects in 13 of 27 studies.

In terms of session duration, short-duration sessions were the most effective, with significant improvements observed in 18 of 23 studies, accounting for 78.3 %. Medium-duration sessions showed effects in 27 of 41 studies, while long-duration sessions were effective in 6 of 11 studies.

For weekly frequency, high-frequency sessions demonstrated significant effects in all 3 studies, representing 100 % (Cho & Roh, 2019; Ji et al., 2017; Seol et al., 2023). Moderate frequency showed significant effects in 32 of 44 studies, corresponding to 72.7 %, while low-frequency sessions were effective in 17 of 28 studies.

With respect to intervention length, long-term interventions yielded the highest efficacy, with significant effects observed in 6 of 8 studies, accounting for 75 %. Mid-term interventions were effective in 20 of 30 studies, while short-term interventions showed significant effects in 26 of 37 studies. The single very long-term intervention included in the analysis did not demonstrate significant effects (Sink et al., 2015).

In terms of exercise intensity, vigorous-intensity exercise produced the most pronounced effects, with significant improvements reported in 10 of 15 studies, representing 66.7 %. Moderate-intensity exercise was effective in 12 of 21 studies, corresponding to 57.1 %. Light- and very light-intensity exercises showed significant effects in one of two studies (Jonasson et al., 2017) and in one study (Byun et al., 2024), respectively.

Additionally, high adherence was the most effective, with significant improvements observed in 17 of 24 studies, accounting for 70.8 %. Moderate adherence was effective in 12 of 18 studies, while low adherence showed no significant effects in either study (Ansai & Rebello, 2015; Merom et al., 2016). These outcomes are summarized in Table 1 and Fig. 3. Furthermore, Fig. 4 illustrates the number of studies for each exercise type across different levels of other exercise parameters.

4. Discussion

4.1. Summary of the results

This systematic review aimed to investigate the effects of chronic exercise interventions on executive function in healthy older adults, focusing on randomized controlled trials. The current literature showed that most studies reported significant beneficial effects of chronic exercise interventions on executive function in this population. Among different types of exercise, aerobic exercise, resistance training, and coordination exercises demonstrated favorable efficacy in improving executive function, with similar proportions of studies reporting significant effects. Additionally, specific exercise parameters, such as short session durations, high frequency, long-term adherence, or vigorous intensity, were associated with improvements in executive function. These findings underline the role of regular, structured exercise as a practical and effective approach to supporting cognitive health and mitigating age-related declines in executive function.

4.2. Discussion on risk of bias

The risk of bias assessment identified several notable trends. Although all studies included in the review were randomized controlled trials, 41 studies lacked detailed descriptions of their randomization processes, leading to “some concerns” ratings for bias due to randomization. This lack of transparency raises issues regarding allocation concealment and its potential impact on outcomes. High dropout rates, particularly among older adults, negatively affected bias ratings for missing outcome data, with 16 studies rated as “some concerns” and another 16 as “high risk”. These findings highlight the difficulty of maintaining participant adherence in this population. Additionally, most studies did not employ or report single-blinding or double-blinding procedures, and some failed to specify exercise parameters such as type, duration, frequency, intensity, or adherence rate. This increased the risk of deviations from intended interventions (“some concerns” in 37 studies, “high risk” in 20 studies) and bias in outcome assessments (“some concerns” in 57 studies). This limitation is especially relevant in trials involving behavioral or exercise interventions, where participant

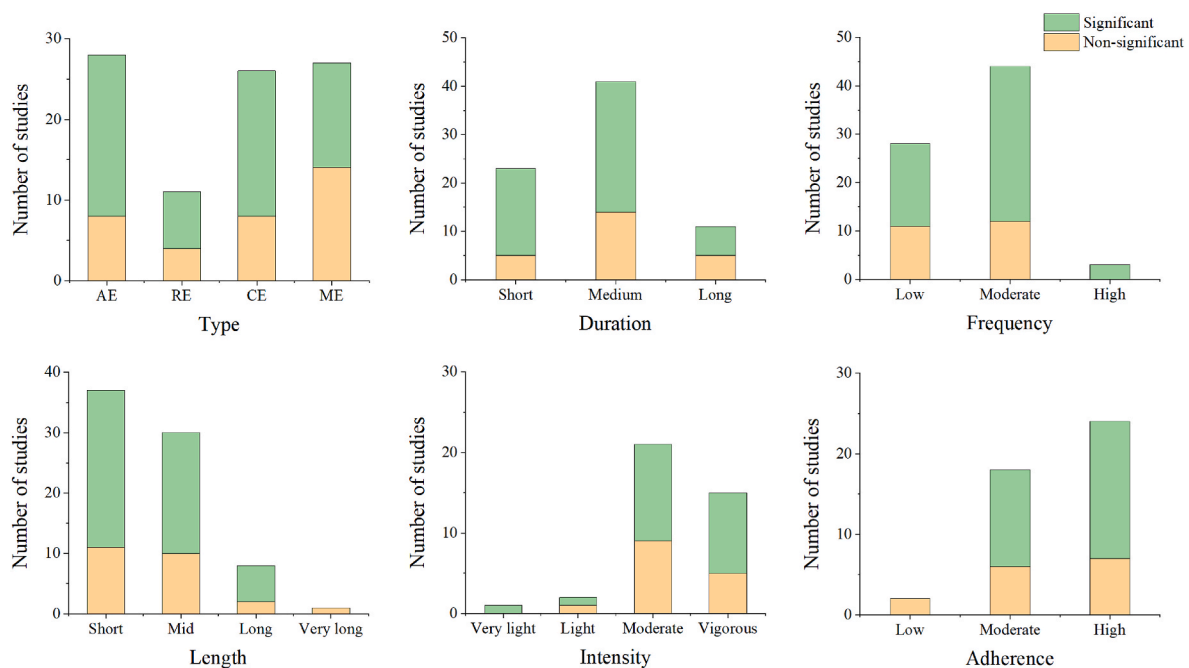


Fig. 3. Number of studies reporting significant and non-significant effects of chronic exercise on executive function across different parameters and adherence levels. Note. AE = aerobic exercise; RE = resistance exercise; CE = coordination exercise; ME = mixed exercise.

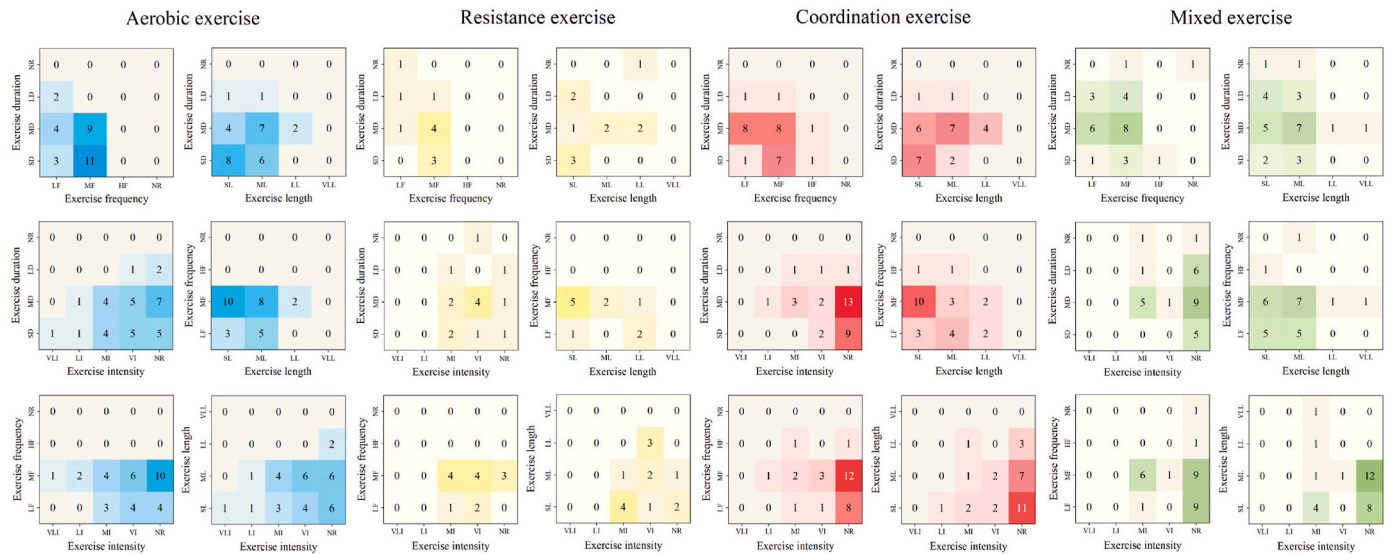


Fig. 4. Illustration showing the number of studies assessing executive function by exercise type, including aerobic exercise, resistance exercise, coordination exercise, and mixed exercise, and associated parameters.

Note. SD = short duration, MD = medium duration, LD = long duration, NR = no reported; LF = low frequency, MF = moderate frequency, HF = high frequency; SL = short length, ML = medium length, LL = long length, VLL = very long length; VLI = very light intensity, LI = light intensity, MI = moderate intensity, VI = vigorous intensity.

and assessor awareness, as well as unreported parameters, may influence results. Furthermore, a few studies did not provide complete baseline or follow-up raw data, leading to “some concerns” in 12 studies and “high risk” in one study, raising concerns about selective reporting bias. While most studies were rated as “low risk” in the selection of reported results, these issues underscore the need for comprehensive data reporting to enhance the reliability and reproducibility of findings.

4.3. Chronic exercise, executive function, and theoretical frameworks

The findings of this systematic review highlight that most studies showed a positive impact of chronic exercise interventions on executive function in healthy older adults. Specifically, 71 % of the included studies (54 out of 76) reported substantial improvements in executive function outcomes, indicating the efficacy of chronic exercise as a non-pharmacological approach for supporting cognitive health and mitigating age-related cognitive decline.

This pattern aligns with previous findings from a systematic review and meta-analysis, which showed that exercise interventions improved multiple aspects of executive function, including working memory, inhibitory control, and cognitive flexibility, in older adults with mild cognitive impairment (Lin et al., 2022). A recent scoping review also indicated that various types of physical exercise, including aerobic, resistance, coordination, and mixed exercises, had positive effects on executive function in older adults (Cai et al., 2024). However, the meta-analysis by Lin et al. (2022) included only 24 studies, and such a small sample size may limit the representativeness and stability of its findings. Additionally, the study focused exclusively on populations with cognitive impairment. Similarly, the scoping review by Cai et al. (2024) included 22 studies involving individuals with mild cognitive impairment, further limiting the generalizability of the findings to cognitively healthy populations. Although another review by Torre and Temprado (2022) examined the effects of exergames on cognitive function in healthy older adults, it reported inconsistent results. Notably, that review included only 23 studies and incorporated non-randomized controlled trials, which may compromise both the internal validity and the reliability of the reported intervention effects due to limited methodological rigor. The current review builds upon these prior studies by including 76 randomized controlled trials and

examining a broader population of healthy older adults. It provides evidence that chronic exercise interventions enhance executive function even in cognitively healthy populations, suggesting that individuals across different health statuses may benefit from such interventions.

These findings are supported by several theoretical frameworks. The Cognitive Reserve Hypothesis posits that the brain can maintain normal function despite age-related changes or pathological damage, such as neurodegenerative diseases, by utilizing neural networks more efficiently or relying on pre-existing cognitive resources (i.e., neural substrates) to support cognitive processes. In contrast, cognitive compensation refers to the brain’s ability to adapt dynamically to deficits by employing alternative strategies or recruiting other brain regions to perform tasks (Stern, 2009). Exercise enhances both cognitive reserve and compensation by improving neural efficiency and adaptability, thereby supporting cognitive function in aging (Cheng, 2016). These mechanisms enable older adults to better manage cognitive demands and delay the onset of clinical symptoms of cognitive decline. Together, cognitive reserve and compensation underscore the vital role of chronic exercise interventions in preserving cognitive health and promoting successful aging.

Additionally, the Adaptive Capacity Model, grounded in aging and neurodegenerative research, offers an evolutionary neuroscience perspective on cognitive health. This model defines adaptive physiological capacity as the neurophysiological system’s ability to adjust dynamically to environmental stimuli. Central to this concept is the evolutionary principle of “use it or lose it” (Raichlen & Alexander, 2017). Exercise is proposed to enhance both bodily and brain adaptation through physiological mechanisms, particularly in activities involving complex environments or high cognitive demands, which require adaptive neural responses to process challenges effectively. In modern society, sedentary lifestyles have led to adaptive downregulation, resulting in decreased brain capacity and contributing to age-related brain atrophy. Exercise counteracts this decline by enhancing neural network flexibility and efficiency, improving cognitive adaptability, and mitigating the negative effects of inactivity. This makes exercise a powerful strategy for maintaining cognitive health during aging (Raichlen & Alexander, 2017). These theoretical perspectives emphasize that exercise is not only a tool for physical health but also a critical intervention for fostering cognitive resilience and adaptability,

underpinning the positive benefits highlighted in this review.

4.4. Types of exercise, executive function, and physiological mechanisms

The type of exercise analysis revealed that aerobic, resistance, and coordination exercises produced similar benefits for executive function, with effect rates ranging from 64 % to 71 %. This consistency suggests that, despite differing pathways (Netz, 2019), these exercise types ultimately enhance executive function. A previous systematic review and meta-analysis examined the effects of physical exercise on executive function in cognitively healthy older adults (Xiong et al., 2021). Although it further analyzed the effects of exercise interventions on different aspects of executive function, it did not investigate the specific influence of exercise parameters such as exercise type. The current study addresses this gap and demonstrates that various types of exercise are all effective in improving executive function. These findings are supported by several review studies. For example, aerobic exercise has been shown to positively enhance executive function subcomponents, including cognitive flexibility, working memory, and inhibitory control, making it a promising intervention for healthy middle-aged and older adults (Ye et al., 2024). Resistance training has also shown positive effects on executive function in older adults, although its impact on memory appears limited (Li et al., 2018). Similarly, coordination exercises have demonstrated beneficial influences on executive function in this population (Cai et al., 2024). While enhancing executive function in older adults is a primary goal of chronic exercise interventions, understanding their underlying neurophysiological mechanisms is equally important.

Aerobic exercise has been shown to increase serum levels of brain-derived neurotrophic factor (BDNF; Griffin et al., 2011), a key regulator of synaptic plasticity and neurogenesis (Numakawa et al., 2018). It also upregulates growth factors such as insulin-like growth factor 1 (IGF-1) and vascular endothelial growth factor (Kang et al., 2020; Vonderwalde & Kovacs-Litman, 2018). IGF-1 contributes to neuroprotection, enhanced neuronal transmission, and regenerative processes (Dyer et al., 2016), while VEGF promotes endothelial cell permeability, proliferation, and angiogenesis (Maharaj & D'Amore, 2007). Similarly, resistance training has been observed to upregulate BDNF and IGF-1 (Herold et al., 2019; Tsai et al., 2019). Additionally, it mitigates mitochondrial dysfunction and inhibits the expression of pro-inflammatory factors in aging cells, thereby countering oxidative and inflammatory processes associated with aging (Cheng et al., 2022). Coordination exercises, on the other hand, may more directly enhance executive function through pathways (e.g., neuromuscular) distinct from those of metabolic exercise (Netz, 2019). These mechanisms are mainly considered to include the promotion of increased neuron and synapse density (Isaacs et al., 1992), the improvement of neural network integration and plasticity (including the sensorimotor cortex, frontal cortex, parietal cortex, basal ganglia, and cerebellum), and the enhancement of neural transmission efficiency (Serrien et al., 2007). These exercise-induced mechanisms may delay or even prevent the onset of neurodegenerative diseases, including Alzheimer's, Parkinson's, and Huntington's diseases (Numakawa et al., 2018).

The relatively lower proportion of studies (48.1 %) reporting significant positive effects of mixed exercise on executive function aligns with previous reviews and meta-analyses in older adults, which have shown smaller effect sizes for combined exercise compared to aerobic, resistance, or coordination exercises alone (Chen et al., 2020). This outcome may be attributed to the varied composition of mixed exercise interventions, which often integrate aerobic, resistance, and coordination components in differing proportions. Such variability can dilute the distinct physiological benefits of individual exercise types, while the concurrent activation of multiple physiological pathways may interfere and fail to produce synergistic effects (Wilson et al., 2012). Additionally, combining different exercise types may complicate implementation, making it challenging to fully deliver the intended benefits of the intervention (Forsyth et al., 2024). Similar inconsistencies are observed

in multimodal interventions such as exergaming and cognitive-motor training. Although previous reviews have reported that these approaches can enhance executive function and overall cognition in healthy older adults (Stojan & Voelcker-Rehage, 2019; Wollesen et al., 2020), results across specific cognitive domains remain inconsistent. Some reported benefits did not persist after sensitivity analyses, and considerable heterogeneity was noted across studies. These findings underscore the need for more rigorous and standardized research protocols. Future interventions should also be better tailored to the specific needs of older adults to more clearly determine the cognitive benefits of mixed exercise approaches.

4.5. Optimal exercise parameters for executive function

A recent review examined the effects of exercise interventions on executive function in older adults (Cai et al., 2024). Although this study evaluated the number of exercise parameters reported across studies, it did not determine which specific parameters were most effective for enhancing executive function. In contrast, the current review provides this critical information by identifying specific exercise parameters associated with the greatest improvements in executive function. Interventions characterized by short session durations (≤ 45 min), high frequency (5–7 sessions per week), long-term duration (25–48 weeks), or vigorous intensity were found to be the most effective. These parameters may maximize benefits on executive function in healthy older adults.

4.5.1. Exercise duration

Short exercise sessions (≤ 45 min) have been shown to positively improve executive function in most studies. Similarly, a previous systematic review and meta-analysis of older adults with mild cognitive impairment reported that short durations had slightly larger effect sizes on executive function compared to medium and long durations (Lin et al., 2022). This suggests that effective exercise training may not require extended durations. Our study extends this finding to healthy older adults and includes an analysis of adherence data, which revealed a higher adherence rate for short durations (85.4 %) compared to medium or long durations (78.2–78.9 %). These results indicate that the greater cognitive benefits observed with shorter durations may be attributed to higher adherence. Past meta-analyses have also highlighted this trend, showing that increased exercise time is associated with a significant decrease in exercise adherence (Burnet et al., 2020). In conclusion, shorter exercise durations may optimize executive function benefits and are associated with higher adherence by reducing time demands and physical strain in older adults.

4.5.2. Exercise frequency

In terms of frequency, high-frequency exercise (5–7 sessions per week) was associated with the greatest improvements in executive function (100 %), although this category included only three studies. Similarly, moderate frequency (3–4 sessions per week) was identified as an optimal parameter in 32 out of 44 studies (72.7 %), suggesting that both moderate and high frequencies have robust effects on executive function in healthy older adults. These findings are consistent with a previous review (Chen et al., 2020), which also highlighted moderate frequency as the most effective, despite ignoring the results of a single high-frequency study. A recent multicenter cross-sectional cohort study found that exercise was positively correlated with cognitive function across all frequencies, with stronger correlations at higher frequencies. Additionally, exercising ≥ 3 times, 4–5 times, or > 5 times per week was significantly negatively correlated with dementia risk, while no significant correlation was found for 1–2 times per week (Wen et al., 2024). These results emphasize the importance of exercise frequency in maintaining cognitive health, suggesting that higher frequencies may provide additional protective benefits against dementia.

4.5.3. Exercise length

Furthermore, long-term interventions (25–48 weeks) were the most effective in enhancing executive function, consistent with findings from previous studies (Chen et al., 2020; Ludyga et al., 2020). These results suggest that extended exercise engagement may facilitate cumulative physiological adaptations and structural brain changes to a greater extent (Dishman et al., 2006), emphasizing the importance of sustained participation in exercise programs to promote cognitive health. These findings suggest optimal parameters and underscore the importance of long-term exercise for improving executive function in older adults. While one very long-term (two-year) study in this research did not show significant benefits, it is important to note that this was a single study with a low adherence rate of 71 %, which may have impacted the results. Therefore, further research is needed to investigate the effects of very long-term exercise interventions on executive function. Improving adherence rates will also be critical to substantiate the dose-response relationship between intervention length and executive function benefits during aging.

4.5.4. Exercise intensity

In terms of intensity, vigorous-intensity exercise demonstrated superior effects on executive function, with 10 out of 15 studies (66.7 %) showing positive results, compared to 12 out of 21 studies (57.1 %) for moderate-intensity exercise or the other intensity levels. These results align with previous systematic reviews and meta-analyses, which have shown that moderate-to-high intensity exercise is effective for improving executive function, with high-intensity exercise yielding larger effect sizes than moderate intensity (Chen et al., 2020). This finding underscores the benefits of moderate-to-high intensity exercise for enhancing executive function in older adults, with high-intensity or vigorous exercise potentially offering more robust benefits by providing more immediate improvements and inducing physiological changes more rapidly (Sanders et al., 2019). However, it is important to note that vigorous exercise may pose challenges for older adults with compromised health, underscoring the need for individualized adjustments based on physical capacity.

4.6. Exercise adherence

Exercise adherence plays a crucial role in determining the effectiveness of interventions (Mahmood et al., 2023). In this review, we found that most studies with high adherence reported significant effects on executive function, in 17 of 24 studies (70.8 %), whereas the two studies with low adherence did not (Ansai & Rebelatto, 2015; Merom et al., 2016). This is reasonable, as low adherence can reduce the effectiveness of an intervention and weaken its intended physiological and cognitive benefits (Evers et al., 2011). These findings suggest that adherence may be a critical factor in the impact of chronic exercise interventions on executive function. However, previous intervention or review studies have largely overlooked the importance of adherence, often failing to report adherence rates or consider them in data analysis (Cai et al., 2024; Wang et al., 2023). This omission increases the risk of deviations from intended interventions and limit the ability to accurately assess the true effects of exercise interventions. Future research should develop adherence strategies at the study planning level and adopt standardized methods to measure and report adherence rate, especially among individuals experiencing multiple barriers such as lack of time, motivation, energy, and environmental support (Room et al., 2017).

4.7. Potential interaction between exercise parameters

Although research on chronic exercise and executive function in older adults is growing (Cai et al., 2024), most studies focus on specific exercise parameters, limiting a comprehensive understanding of their interactions. Fig. 4 highlights the distribution of studies examining these

interactions, guiding future research to address underexplored parameter combinations. Resistance training remains less studied than other exercise types, despite its known benefits for executive function (Li et al., 2018). More research is needed to optimize exercise prescriptions by exploring parameter interactions. Specifically, most studies primarily focus on interventions of medium duration (46–60 min), moderate frequency (3–4 sessions per week), short-term duration (≤ 12 weeks), and moderate intensity. However, the effects of other parameter combinations, such as longer durations (≥ 61 min) with high frequency (5–7 sessions per week), very long-term interventions (≥ 49 weeks) with light intensity, longer durations combined with very long-term interventions, or high frequency with light intensity, remain largely unexplored. Additionally, exercise intensity is often underreported, increasing the risk of deviations from the intended intervention. Future research should prioritize underexplored combinations of exercise parameters, particularly in resistance training, to refine exercise prescriptions for executive function enhancement in older adults.

4.8. Limitations of the current study

Despite this review's valuable contributions in summarizing the current literature and providing insights, several limitations must be acknowledged. First, comparing individual exercise parameters has inherent limitations. Analyzing each parameter of exercise intervention separately may lead to oversimplified conclusions that fail to capture the potential interactions effects between exercise parameters on executive function and therefore to identify optimal parameters. Although this approach remains common due to the limited number of studies, the current review provided an initial attempt to describe the parametric interactions of exercise interventions and their relations to the percentage of showing beneficial effects on executive function. These preliminary findings suggest the need to conduct more studies manipulating resistance exercise, long-duration interventions (≥ 61 min), high-frequency protocols (5–7 sessions per week), very long-term interventions (≥ 49 weeks), and light-intensity exercise.

Second, accurately classifying exercise types remains a significant challenge. Exercise types were categorized based on those reported in the original studies, yet most exercises include varying proportions of multiple components. For example, resistance training may simultaneously involve components of balance or flexibility. This lack of precision in isolating a single exercise type may compromise the internal validity of comparisons. To address this, future studies should develop more detailed intervention plans and implementation strategies that account for potential confounding factors, especially when exercises involve multiple components. Carefully considering these factors and minimizing their impact will help ensure more accurate and reliable comparisons. Additionally, it is recommended that future research enhance transparency by clearly reporting study details, allowing others to replicate the same protocols and compare their findings.

Finally, a meta-analytic approach was not pursued in this study. The included studies exhibited considerable variability in both exercise parameters (e.g., type, duration, frequency, length, intensity) and the specific executive function domains assessed (e.g., working memory, inhibitory control, cognitive flexibility). Such methodological heterogeneity limits the feasibility and validity of conducting a pooled meta-analysis, as combining these diverse interventions and outcomes may result in misleading or uninterpretable findings. Instead, a narrative synthesis was employed to allow for a comprehensive and nuanced examination of patterns and trends across studies, offering a different yet valuable perspective. While this method does not provide effect size estimates, it remains appropriate and informative in the current context, offering meaningful insights to guide the development of more targeted and theory-driven exercise interventions aimed at enhancing executive function in healthy older adults.

5. Conclusion

This systematic review demonstrates that chronic exercise interventions can enhance executive function in healthy older adults. Similar improvements were observed across aerobic, resistance, and coordination exercise interventions. The findings provide preliminary evidence for optimal intervention parameters, such as short durations, high frequency, long-term engagement, and vigorous intensity, offering valuable guidance for designing effective exercise programs. Additionally, the review suggests that improving adherence may enhance intervention effectiveness. Furthermore, the study emphasizes the importance of exploring interactions between exercise parameters, identifying research gaps, and guiding future studies. By presenting these insights, this review underscores the role of structured exercise in mitigating age-related cognitive decline and serves as a valuable reference for developing strategies to promote healthy aging.

CRedit authorship contribution statement

Chih-Chien Lin: Writing – original draft, Visualization, Formal analysis, Data curation, Conceptualization. **Shih-Chun Kao:** Writing – review & editing, Methodology, Conceptualization. **Ting-Yu Chueh:** Validation, Investigation. **Tsung-Min Hung:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

This study received no financial or non-financial support. The authors declare no conflicts of interest related to the content or publication of this study.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.psychsport.2025.102925>.

Data availability

No data was used for the research described in the article.

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